The Stieltjes transform

Def: Let μ be a probability measure on $\mathbb{R}$, let $\mathrm{supp}(\mu)$ be its (closed) support. The Stieltjes transform of μ is the function $m: \mathbb{C} \setminus \mathrm{supp}(\mu) \to \mathbb{C}$

$$m(z) = \int \frac{1}{x - z} \, d\mu(x) = \mathop{\mathbb{E}}_{X \sim \mu} \left[\frac{1}{X - z} \right]$$

[Note: Some authors use $\int \frac{1}{z - x} \, d\mu(x)$]

If μ_n is the ESD of a symmetric matrix $X \in \mathbb{R}^{n \times n}$, then its Stieltjes transform is

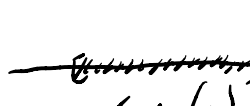
$$m_n(z) = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{\lambda_i - z} = \frac{1}{n} \mathrm{Tr} \left(X - zI \right)^{-1}$$

The matrix $R(z) = (X - zI)^{-1}$ is the resolvent of X.

Properties of the Stieltjes transform:

① $\left| \frac{1}{x - z} \right| = \frac{1}{|x - z|} \leq \frac{1}{\mathrm{dist}(z, \mathrm{supp}(\mu))} \quad \forall x \in \mathrm{supp}(\mu)$

Thus: $|m(z)| \leq \int \left| \frac{1}{x - z} \right| d\mu(x) \leq \frac{1}{\mathrm{dist}(z, \mathrm{supp}(\mu))}$



$m(z)$ is locally bounded ($\forall z \in \mathbb{C} \setminus \mathrm{supp}(\mu)$, there is an open nbd. $U \ni z$ where $m(z)$ is bounded).

(2) Similarly $\left|\frac{d}{dz}\frac{1}{x-z}\right| = \left|\frac{1}{(x-z)^2}\right| = \frac{1}{|x-z|^2} \le \frac{1}{\mathrm{dist}(z, \mathrm{supp}(\mu))^2}$

By dominated convergence: m is (complex) differentiable,

$$m'(z) = \int \frac{d}{dz}\frac{1}{x-z}\, d\mu(x) = \int \frac{1}{(x-z)^2}\, d\mu(x)$$

and $|m'(z)| \le \frac{1}{\mathrm{dist}(z, \mathrm{supp}(\mu))^2}$ so $m(z)$ is <u>locally Lipschitz</u>

($\forall z \in \mathbb{C} \setminus \mathrm{supp}(\mu)$, there is open nbd. $\ni z$ where $m(z)$ is Lipschitz)

(3) Suppose μ has compact support, $\mathrm{supp}(\mu) \subseteq [-R, R]$.

For $z \in \mathbb{C}$ w/ $|z| > R$:

$$m(z) = \int \frac{1}{x-z}\, d\mu(x)$$

$$= \int -\frac{1}{z}\left(\frac{1}{1-\frac{x}{z}}\right) d\mu(x)$$

$$= \int -\frac{1}{z}\left(1 + \frac{x}{z} + \frac{x^2}{z^2} + \frac{x^3}{z^3} + \cdots\right) d\mu(x)$$

$$= -\sum_{k=0}^{\infty} \frac{m_k}{z^{k+1}}, \qquad m_k := \int x^k\, d\mu(x) \text{ } k^{\text{th}} \text{ moment of } \mu.$$

where all series are absolutely convergent.

In particular, $m(z) = -\frac{1}{z} + O\left(\frac{1}{|z|^2}\right)$ for large z.

④ Let $z = \lambda + i\eta$ ($\lambda = \mathrm{Re}\, z$, $\eta = \mathrm{Im}\, z$). Then

$$m(z) = \int \frac{1}{x - \lambda - i\eta}\, d\mu(x) = \int \frac{x - \lambda + i\eta}{(x-\lambda)^2 + \eta^2}\, d\mu(x)$$

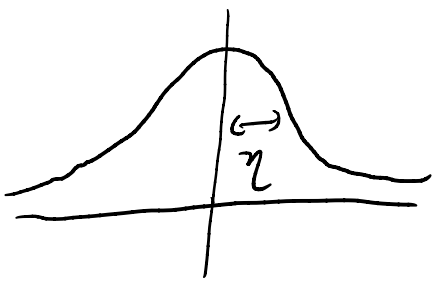
$$\Rightarrow \begin{cases} \mathrm{Re}\, m(z) = \displaystyle\int \frac{x-\lambda}{(x-\lambda)^2+\eta^2}\, d\mu(x) \\ \mathrm{Im}\, m(z) = \displaystyle\int \frac{\eta}{(x-\lambda)^2+\eta^2}\, d\mu(x) \end{cases}$$

If $z \in \mathbb{C}^+$ ($\eta > 0$) then $m(z) \in \mathbb{C}^+$. Also

$$m(\bar{z}) = \int \frac{1}{x - \bar{z}}\, d\mu(x) = \int \overline{\left(\frac{1}{x-z}\right)}\, d\mu(x) = \overline{m(z)}.$$

So $m(z)$ is conjugate-symmetric, often defined as $m : \mathbb{C}^+ \to \mathbb{C}^+$.

⑤ Let $p_\eta(x) = \frac{1}{\pi} \cdot \frac{\eta}{x^2 + \eta^2}$

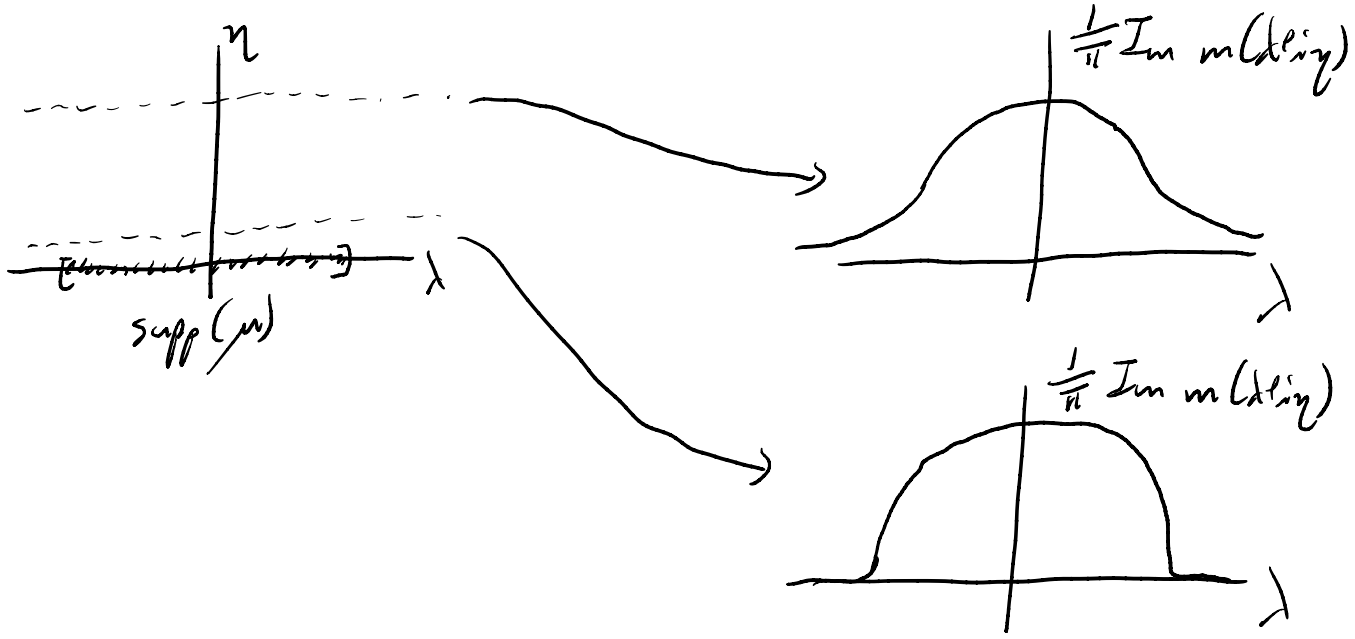


be the Cauchy$(0, \eta)$ density

Then fixing $\eta = \mathrm{Im}\, z$,

$$\lambda \mapsto \frac{1}{\pi} \mathrm{Im}\, m(z) = \int p_\eta(\lambda - x)\, d\mu(x)$$

is the density of the convolution $\mu * \mathrm{Cauchy}(0, \eta)$:



Prop (Stieltjes inversion): For any $x \in \mathbb{R}$ where $F(x) := \mu((-\infty, x])$ is continuous, $F(x) = \lim_{\eta \to 0} \int_{-\infty}^{x} \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta)\, d\lambda$.

If $f(x) = \lim_{z \in \mathbb{C}^+ \to x} \frac{1}{\pi} \operatorname{Im} m(z)$ exists, then μ has density $f(x) = F'(x)$ at x.

In particular: $\{m(z) : z \in \mathbb{C}^+\}$ uniquely determines μ.

Proof: Let $X \sim \mu$ and $Z_\eta \sim \text{Cauchy}(0, \eta)$ be independent.

Then $\int_{-\infty}^{x} \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta)\, d\lambda = \mathbb{P}[X + Z_\eta \le x]$.

As $\eta \to 0$, $Z_\eta \to 0$ in probability, so for cont. point x of $F(x)$,

$$\mathbb{P}[X + Z_\eta \le x] \to \mathbb{P}[X \le x] = F(x).$$

For second claim: Suppose $f(x) = \lim_{z \in \mathbb{C}^+ \to x} \frac{1}{\pi} \operatorname{Im} m(z)$ exists.

Suffices to show

$$f(x) = \lim_{x_k \nearrow x,\, y_k \searrow x} \frac{F(y_k) - F(x_k)}{y_k - x_k}$$

along any sequences $x_k \nearrow x$ and $y_k \searrow x$ of cont. points of F.

For any $\varepsilon > 0$:

- For each $k \in \mathbb{N}$, by first claim, $\exists\, \eta_k > 0$ s.t.

$$\left| \int_{x_k}^{y_k} \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta_k)\, d\lambda - (F(y_k) - F(x_k)) \right| < \varepsilon (y_k - x_k)$$

- For all large enough $k \in \mathbb{N}$, by existence of $f(x)$,

$$\left| \frac{1}{y_k - x_k} \int_{x_k}^{y_k} \frac{1}{\pi} \operatorname{Im} m(d + i\eta_k) - f(x) \right| < \varepsilon$$

$$\Rightarrow \limsup_{k \to \infty} \left| \frac{F(y_k) - F(x_k)}{y_k - x_k} - f(x) \right| < 2\varepsilon. \quad \text{Take } \varepsilon \to 0. \quad \blacksquare$$

Prop (characterization of weak convergence): Suppose $\{m_n(z)\}_{n=1}^{\infty}$, $m(z)$ are the Stieltjes transforms of probability measures μ_n, μ, and

$$\lim_{n \to \infty} m_n(z) = m(z) \quad \forall z \in \mathbb{C}^+.$$

Then $\mu_n \to \mu$ weakly.

Proof: Let $f \in C_0(\mathbb{R})$ (cont., bdd., compact support).

For any $\eta > 0$, by dominated convergence,

$$\lim_{n\to\infty} \int_{\mathbb{R}} f(\lambda) \cdot \frac{1}{\pi} \operatorname{Im} m_n(\lambda + i\eta)\, d\lambda = \int_{\mathbb{R}} f(\lambda) \cdot \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta)\, d\lambda.$$

We have $\int_{\mathbb{R}} f(\lambda) \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta)\, d\lambda$

$$= \int_{\mathbb{R}} f(\lambda) \cdot \int \frac{1}{\pi} \cdot \frac{\eta}{(\lambda - x)^2 + \eta^2}\, d\mu(x)\, d\lambda$$

$$= \int \left[\int f(\lambda) \cdot \frac{1}{\pi} \frac{\eta}{(\lambda - x)^2 + \eta^2}\, d\lambda \right] d\mu(x)$$

Hence $\left| f(x) - \int f(\lambda) \frac{1}{\pi} \frac{\eta}{(\lambda - x)^2 + \eta^2}\, d\lambda \right|$

$$= \left| \int \left(f(x) - f(\lambda + x) \right) \cdot \frac{1}{\pi} \frac{\eta}{\lambda^2 + \eta^2}\, d\lambda \right|$$

$$= \int_{|\lambda| \le \delta} + \int_{|\lambda| > \delta}$$

$$\le \underbrace{\sup_{x \in \mathbb{R},\, |\lambda| \le \delta} |f(x) - f(\lambda + x)|}_{:= V_\delta(f)} + 2\|f\|_\infty \mathbb{P}[|Z_\eta| > \delta].$$

So $\left| \int_{\mathbb{R}} f(\lambda) \cdot \frac{1}{\pi} \operatorname{Im} m(\lambda + i\eta)\, d\lambda - \int f(x)\, d\mu(x) \right|$

$$\le V_\delta(f) + 2\|f\|_\infty \mathbb{P}[|Z_\eta| > \delta].$$

Same bound holds for m_n, μ_n (uniformly in n).

$\Rightarrow \limsup_{n\to\infty} \left| \int f(x)\, d\mu_n(x) - \int f(x)\, d\mu(x) \right| \le 2\left(V_\delta(f) + 2\|f\|_\infty \mathbb{P}[|Z_n| > \delta]\right)$.

Take $\eta \to 0$, then $\delta \to 0$, note $V_\delta(f) \to 0$ since f must be uniformly continuous $\Rightarrow \lim_{n\to\infty} \int f\, d\mu_n = \int f\, d\mu \quad \forall f \in C_0(\mathbb{R})$.

For any continuous bdd. $f: \mathbb{R} \to \mathbb{R}$, any $\varepsilon > 0$:

Let $\phi \in C_0(\mathbb{R})$, $0 \le \phi \le 1$, $\int \phi\, d\mu \ge 1 - \varepsilon$. Then

$$\int f\phi\, d\mu_n \to \int f\phi\, d\mu \text{ since } f\phi \in C_0(\mathbb{R}),$$

$$\left|\int f(1-\phi)\, d\mu\right| \le \|f\|_\infty \cdot \varepsilon, \quad \limsup_{n\to\infty} \left|\int f(1-\phi)\, d\mu_n\right| \le \|f\|_\infty \cdot \varepsilon.$$

$$\Rightarrow \limsup_{n\to\infty} \left|\int f\, d\mu_n - \int f\, d\mu\right| \le 2\|f\|_\infty \cdot \varepsilon.$$

Take $\varepsilon \to 0$: $\int f\, d\mu_n \to \int f\, d\mu$. ∎

(6) $m(z)$ is complex analytic on $\mathbb{C}^+$. Then for any countable $S \subset \mathbb{C}^+$ w/ accumulation point in $\mathbb{C}^+$:

- If m_1, m_2 are Stieltjes transf. of μ_1, μ_2 s.t. $m_1(z) = m_2(z)$ $\forall z \in S$, then also $m_1(z) = m_2(z)$ $\forall z \in \mathbb{C}^+$, hence $\mu_1 = \mu_2$.
- If m_n, m are Stieltjes transf. of μ_n, μ s.t. $\lim_{n\to\infty} m_n(z) = m(z)$ $\forall z \in S$, then also $\lim_{n\to\infty} m_n(z) = m(z)$ $\forall z \in \mathbb{C}^+$, hence $\mu_n \to \mu$ weakly.

Cor: If μ_n are random probability measures on $\mathbb{R}$ s.t. for each fixed $z \in \mathbb{C}^+$, $m_n(z) \to m(z)$ a.s., then $\mu_n \to \mu$ weakly a.s.

Wigner's semicircle law via Stieltjes transform

Let $X \in \mathbb{R}^{n \times n}$ be Wigner matrix, μ_n its ESD.

Heuristic: $m_n(z) = \frac{1}{n} \operatorname{Tr}(X - zI)^{-1} = \frac{1}{n} \sum_{i=1}^{n} \left[(X - zI)^{-1}\right]_{ii}$.

Recall Schur-complement $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} \underbrace{(A - BD^{-1}C)^{-1}}_{:=S} & -SBD^{-1} \\ -D^{-1}CS & D^{-1} + D^{-1}CSBD^{-1} \end{pmatrix}$

$$\Rightarrow \left[(X - zI)^{-1}\right]_{ii} = \frac{1}{(X_{ii} - z) - X_i^T (X^{(i)} - zI)^{-1} X_i} \quad \text{where}$$

- X_i = i^{th} column of X w/ i^{th} entry removed $\in \mathbb{R}^{n-1}$
- $X^{(i)}$ = X w/ i^{th} row + i^{th} column removed $\in \mathbb{R}^{(n-1)\times(n-1)}$

Importantly, $X^{(i)}$ is independent of X_i. Apply:

- $X_{ii} \approx 0$
- $X_i^T A X_i \approx \mathbb{E}\, X_i^T A X_i = \mathbb{E} \sum_{j=1}^{n} X_{ij}^2 a_{jj} = \frac{1}{n} \operatorname{Tr} A$ for deterministic A
- $\frac{1}{n} \operatorname{Tr}(X^{(i)} - zI)^{-1} \approx \frac{1}{n} \operatorname{Tr}(X - zI)^{-1} = m_n(z)$.

$$\Rightarrow m_n(z) = \frac{1}{n} \sum_{i=1}^{n} \left[(X - zI)^{-1}\right]_{ii} \approx \frac{1}{-z - m_n(z)}.$$

For $z \in \mathbb{C}^+$, as $n \to \infty$, we expect $m_n(z) \to m_{sc}(z) \in \mathbb{C}^+$

solving $m_{sc}(z) = \frac{1}{-z - m_{sc}(z)}$, i.e. $m_{sc}(z)^2 + z\, m_{sc}(z) + 1 = 0$.

$$\Rightarrow m_{sc}(z) = \frac{-z + \sqrt{z^2 - 4}}{2}$$ choosing $\sqrt{\cdot}$ w/ positive imaginary part.

For $x \in \mathbb{R}$, $\lim_{z \in \mathbb{C}^+ \to x} m_{sc}(z) = \begin{cases} \frac{-x + \sqrt{x^2-4}}{2} & \text{if } x \notin [-2,2] \\ \frac{-x + i\sqrt{4-x^2}}{2} & \text{if } x \in [-2,2] \end{cases}$

$$\Rightarrow f(x) = \lim_{z \in \mathbb{C}^+ \to x} \frac{1}{\pi} \operatorname{Im} m_{sc}(z) = \frac{1}{2\pi} \sqrt{4-x^2}\, \mathbb{1}_{[-2,2]}(x)$$ — semicircle law!

Lemma: Let $z \in \mathbb{R}^n$ have independent entries w/ $\mathbb{E} z_i = 0$, $\mathbb{E} z_i^2 = 1$, $\mathbb{E}|z_i|^k \le C_k$ for some constant $C_k > 0$ and each $k \ge 1$. Then $\exists$ constants $C_1', C_2', C_3', \ldots > 0$ s.t. for any $p \ge 1$:

(a) $\mathbb{P}[|u^T z| \ge t] \le C_p' \left(\frac{\|u\|_2}{t}\right)^{2p}$ $\forall$ deterministic $u \in \mathbb{R}^n$, $t > 0$

(b) $\mathbb{P}[|z^T A z - \operatorname{Tr} A| \ge t] \le C_p' \left(\frac{\|A\|_F}{t}\right)^{2p}$

$\forall$ deterministic $A \in \mathbb{R}^{n \times n}$, $t > 0$.

Proof: (a) For any $p \ge 1$, $\mathbb{E}(u^T z)^{2p} = \sum_{i_1, \ldots, i_{2p}=1}^{n} \underbrace{\mathbb{E}\, u_{i_1} \cdots u_{i_{2p}} z_{i_1} \cdots z_{i_{2p}}}_{(*)}$

- If any $i_1, \ldots, i_{2p}$ is unique, then $(*) = 0$ b/c $\mathbb{E} z_i = 0$.
- Each other $(i_1, \ldots, i_{2p})$ induces a partition $\Pi(i_1, \ldots, i_{2p})$ of $\{1, \ldots, 2p\}$

into blocks of indices that are equal, of sizes $k_1, \dots, k_r \geq 2$.

Fixing any such partition Π and bounding $\mathbb{E}|z_{i_1} \cdots z_{i_{2p}}| \leq C_p'$,

$$\sum_{\substack{i_1,\dots,i_{2p} \\ \Pi(i_1,\dots,i_{2p})=\Pi}} |(*)| \leq C_p' \sum_{j_1,\dots,j_r=1}^{n} |u_{j_1}|^{k_1} \cdots |u_{j_r}|^{k_r}$$

$$\leq C_p' \|u\|_\infty^{(k_1-2)+\cdots+(k_r-2)} \sum_{j_1,\dots,j_r=1}^{n} |u_{j_1}|^2 \cdots |u_{j_r}|^2$$

$$\leq C_p' \|u\|_2^{k_1+\cdots+k_r} = C_p' \|u\|_2^{2p}$$

Number of partitions is at most some $C_p'' > 0$. Thus

$$\mathbb{E}(u^T z)^{2p} \leq C_p' C_p'' \|u\|_2^{2p} \Rightarrow \mathbb{P}[|u^T z| \geq t] \leq \frac{C_p' C_p'' \|u\|_2^{2p}}{t^{2p}}.$$

(b) $$z^T A z - \operatorname{Tr} A = \underbrace{\sum_{i=1}^{n} (z_i^2 - 1) a_{ii}}_{(I)} + \underbrace{\sum_{i \neq j} z_i z_j a_{ij}}_{(II)}.$$

By (a), $\mathbb{E}(I)^{2p} \leq C_p' \|(a_{11}, \dots, a_{nn})\|_2^{2p} \leq C_p' \|A\|_F^{2p}$.

For (II): Let S contain each $i \in \{1, \dots, n\}$ independently w/ prob. $1/2$.

Note $$\mathbb{E}_S \sum_{\substack{i \in S \\ j \notin S}} z_i z_j a_{ij} = \mathbb{E}_S \sum_{i \neq j} z_i z_j a_{ij} \mathbb{1}\{i \in S, j \notin S\} = \frac{1}{4}(II).$$

Then $$\mathbb{E}(II)^{2p} = \mathbb{E}_z \left(4 \mathbb{E}_S \sum_{\substack{i \in S \\ j \notin S}} z_i z_j a_{ij} \right)^{2p}$$

$$\overset{\text{(Jensen's)}}{\leq} 4^{2p}\, \mathbb{E}_{z,S}\left(\sum_{\substack{i\in S\\ j\notin S}} z_i z_j a_{ij}\right)^{2p}.$$

Fix S, set $u_j = \sum_{i\in S} z_i a_{ij}$. Then

$$\mathbb{E}_z\left(\sum_{\substack{i\in S\\ j\notin S}} z_i z_j a_{ij}\right)^{2p} = \mathbb{E}_z\left(\sum_{j\notin S} u_j z_j\right)^{2p}$$

$$\overset{(a)}{\leq} C_p'\, \mathbb{E}_z\left(\sum_{j\notin S} u_j^2\right)^p := C_p' \left\|\sum_{j\notin S} u_j^2\right\|_{L^p}^p$$

$$\leq C_p'\left(\sum_{j\notin S} \|u_j^2\|_{L^p}\right)^p \quad (L^p\text{-triangle ineq.})$$

$$= C_p'\left(\sum_{j\notin S} \left(\mathbb{E}\, u_j^{2p}\right)^{1/p}\right)^p$$

$$\overset{(a)}{\leq} C_p'\left(\sum_{j\notin S} C_p' \sum_{i\in S} a_{ij}^2\right)^p \leq C_p'' \|A\|_F^{2p}.$$

So $\mathbb{E}\,|z^TAz - \mathrm{Tr}\,A|^{2p} \leq \mathbb{E}\,[(I)+(II)]^{2p} \leq C_p'''\|A\|_F^{2p}$

$$\Rightarrow \mathbb{P}\left[|z^TAz - \mathrm{Tr}\,A| \geq t\right] \leq \frac{C_p'''\|A\|_F^{2p}}{t^{2p}}.$$

∎

Lemma: For any symmetric matrix $X\in\mathbb{R}^{n\times n}$ and $z=\lambda+i\eta\in\mathbb{C}^+$,

$$\left|\mathrm{Tr}\,(X-zI)^{-1} - \mathrm{Tr}\,(X^{(i)}-zI)^{-1}\right| \leq \frac{\pi}{\eta}$$

Proof: Let $\lambda_1 \leq \dots \leq \lambda_n$ be eig. values of X,

$\mu_1 \leq \dots \leq \mu_{n-1}$ be eig. values of $X^{(i)}$

By Cauchy-interlacing,

$$\lambda_1 \quad \lambda_2 \quad \cdots \quad \lambda_{n-1} \quad \lambda_n$$
$$\mu_1 \quad \mu_2 \quad \cdots \quad \mu_{n-1}$$

Let $F(x) = \sum_{i=1}^{n} \mathbb{1}_{(-\infty, x]}(\lambda_i)$, $F^{(i)}(x) = \sum_{i=1}^{n-1} \mathbb{1}_{(-\infty, x]}(\mu_i)$.

Cauchy-interlacing implies $|F(x) - F^{(i)}(x)| \le 1 \quad \forall x \in \mathbb{R}$.

$$\left| \mathrm{Tr}\,(X - zI)^{-1} - \mathrm{Tr}\,(X^{(i)} - zI)^{-1} \right|$$

$$= \left| \int \frac{1}{x - z}\, dF(x) - \int \frac{1}{x - z}\, dF^{(i)}(x) \right|$$

$$= \left| \int \frac{F(x)}{(x-z)^2}\, dx - \int \frac{F^{(i)}(x)}{(x-z)^2}\, dx \right| \quad \text{(Riemann-Stieltjes integ. by parts)}$$

$$\le \int \frac{1}{|x - z|^2}\, dx = \int \frac{1}{(x-\lambda)^2 + \eta^2}\, dx = \frac{\pi}{\eta}. \qquad \blacksquare$$

Proof #2 of Wigner's Thm: By Schur-complement,

$$m_n(z) = \frac{1}{n} \mathrm{Tr}\,(X - zI)^{-1} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{x_{ii} - z - x_i^T \underbrace{(X^{(i)} - zI)^{-1}}_{:= R^{(i)}(z)} x_i}$$

- Recall $x_i = \left(\frac{1}{\sqrt{n}} z_{ij}\right)_{j \ne i}$. Applying Lemma #1 conditional on $X^{(i)}$, w/ $A = \mathrm{Re}\, \frac{1}{n} R^{(i)}(z)$ and $A = \frac{1}{n} \mathrm{Im}\, R^{(i)}(z)$ (entrywise Re/Im):

$$\mathbb{P}\left[\left| x_i^T R^{(i)}(z) x_i - \frac{1}{n} \mathrm{Tr}\, R^{(i)}(z) \right| \ge t \,\middle|\, X^{(i)} \right] \le C_p \left(\frac{\left\| \frac{1}{n} R^{(i)}(z) \right\|_F}{t} \right)^{2p}$$

For any symmetric $X = V\begin{pmatrix}\lambda_1 & & \\ & \ddots & \\ & & \lambda_n\end{pmatrix}V^T$, $z = \lambda + i\eta \in \mathbb{C}^+$

$$\|(X - zI)^{-1}\|_F^2 = \operatorname{Tr}\,(X-zI)^{-1}\overline{(X-zI)^{-1}}^*$$

$$= \operatorname{Tr}\, V\begin{pmatrix}\frac{1}{\lambda_1 - z} & & \\ & \ddots & \\ & & \frac{1}{\lambda_n - z}\end{pmatrix}V^T V\begin{pmatrix}\frac{1}{\lambda_1 - \bar{z}} & & \\ & \ddots & \\ & & \frac{1}{\lambda_n - \bar{z}}\end{pmatrix}V$$

$$= \sum_{i=1}^n \frac{1}{\lambda_i - z}\cdot\frac{1}{\lambda_i - \bar{z}} = \sum_{i=1}^n \frac{1}{|\lambda_i - z|^2} \le n\cdot\frac{1}{\eta^2}$$

$$\Rightarrow \left\|\frac{1}{n}R^{(i)}(z)\right\|_F \le \frac{1}{\eta\sqrt{n}}.$$ Fix any $t>0$, $p \ge 2$, take union bound over $i = 1, \dots, n$ + Borel-Cantelli:

$$\sup_{i=1}^n \left| X_i^T R^{(i)}(z) X_i - \frac{1}{n}\operatorname{Tr} R^{(i)}(z)\right| \to 0 \text{ a.s. as } n\to\infty.$$

- Applying Lemma #2:

$$\sup_{i=1}^n \left|\frac{1}{n}\operatorname{Tr} R^{(i)}(z) - \underbrace{\frac{1}{n}\operatorname{Tr} R(z)}_{=m_n(z)}\right| \le \frac{\pi}{n\eta} \to 0 \text{ as } n\to\infty.$$

- Also $\mathbb{P}[|X_{ii}| > t] = \mathbb{P}[|Z_{ii}| > t\sqrt{n}] \le \frac{\mathbb{E}|Z_{ii}|^p}{(t\sqrt{n})^p} \le \frac{C_p}{(t\sqrt{n})^p}$.

Fix any $t>0$, $p \ge 2$, take union bound + Borel-Cantelli:

$$\sup_{i=1}^n |X_{ii}| \to 0 \text{ a.s. as } n\to\infty.$$

Thus: For some $r_n^{(i)}$ satisfying $\sup_{i=1}^n |r_n^{(i)}| \to 0$ a.s.:

$$m_n(z) = \frac{1}{n}\sum_{i=1}^n \frac{1}{-z - m_n(z) + r_n^{(i)}}.$$

Note: $|z + m_n(z)| \ge \operatorname{Im}(z + m_n(z)) \ge \operatorname{Im} z = \eta$. Thus

$$\left| \frac{1}{-z - m_n(z) + r_n^{(i)}} - \frac{1}{-z - m_n(z)} \right| \leq \frac{|r_n^{(i)}|}{\eta(\eta - |r_n^{(i)}|)}.$$

So $m_n(z) = \frac{1}{-z - m_n(z)} + r_n$, where $|r_n| \to 0$ a.s.

$$\Rightarrow m_n(z)^2 + (z + r_n) m_n(z) + (1 - r_n z) = 0.$$

$$\Rightarrow m_n(z) \to m_{sc}(z) = \frac{-z + \sqrt{z^2 - 4}}{2} \text{ a.s. } \Rightarrow \mu_n \to \mu \text{ weakly a.s. } \blacksquare$$

Problem 3: Let $X \in \mathbb{R}^{n \times n}$ be a Wigner matrix, $R(z) = (X - zI)^{-1}$ its resolvent. Fixing $z \in \mathbb{C}^+$, show that a.s. as $n \to \infty$:

(a) $\sup_{i=1}^{n} |R(z)_{ii} - m_{sc}(z)| \to 0.$

(b) $\sup_{i \neq j} |R(z)_{ij}| \to 0.$

(Hint: Show via Schur-complement identity that

$$R(z)_{ij} = -R(z)_{ii} [R^{(i)}(z) x_i]_j \;)$$

(c) Let $A \in \mathbb{R}^{n \times n}$ be any matrix s.t. $\|A\|_{op} \leq C$ for a constant $C > 0$ (not depending on n). Then

$$\frac{1}{n} \operatorname{Tr} A[R(z) - m_{sc}(z) \cdot I] \to 0.$$